AV Tech Network Router Guide

TP-Link ER605 Router, WireGuard VPN, and PoE Camera Switch

New Hope Lutheran Church - Stream Agent III sd AV System

Prepared May 2026

Internal-use warning: This document includes internal network addresses and the ER605 login. Keep it with the internal AV tech documentation and do not post it publicly.

Contents

1. Why the ER605 matters to the AV system

The TP-Link ER605 is the router/gateway for the church AV technology network. It is the device that makes the 192.168.88.x AV LAN work as a separate, controlled network behind the main SaskTel/Arris internet gateway.

- It provides the AV LAN gateway at 192.168.88.1.
- It assigns DHCP addresses to AV devices such as BeelinkOBS, the Raspberry Pi WOL helper, tablet clients, and the access point when those devices are not configured with static addresses.
- It routes traffic from the AV LAN out to the church internet connection for YouTube streaming and updates.
- It hosts the WireGuard VPN server used for remote troubleshooting and remote AV support.
- It keeps the camera/control network organized so OBS, Stream Agent III sd, Master Fader, Proclaim, and camera control can work reliably together.

In this design, the ER605 does not power cameras. Power over Ethernet (PoE) is handled by the separate TP-Link TL-SG1005P PoE switch. This separation is deliberate: the router handles routing/VPN/DHCP, while the PoE switch handles camera power and camera data.

2. Why the MikroTik router was replaced

The previous MikroTik PoE router was important in the earlier system because it supplied the camera/control subnet and also powered cameras through PoE. It is no longer trusted for the current AV system.

- The MikroTik power supply was believed to be degraded and no longer able to supply enough current at the correct 48 V level for the cameras.
- Because the same device was also responsible for data switching/routing, the power problem raised concern that transmitting and receiving data could also be compromised.
- The MikroTik system failed during a church service. Troubleshooting during the service was not practical without disrupting the entire streaming setup.
- The replacement design separates routing from PoE power, which makes failures easier to isolate and makes the equipment easier to access in the AV tech cabinet.

Important distinction: This is not a general criticism of MikroTik as a brand. The decision is based on the reliability concerns with the specific church unit and its power/PoE role in the live streaming chain.

3. Network layout and device roles

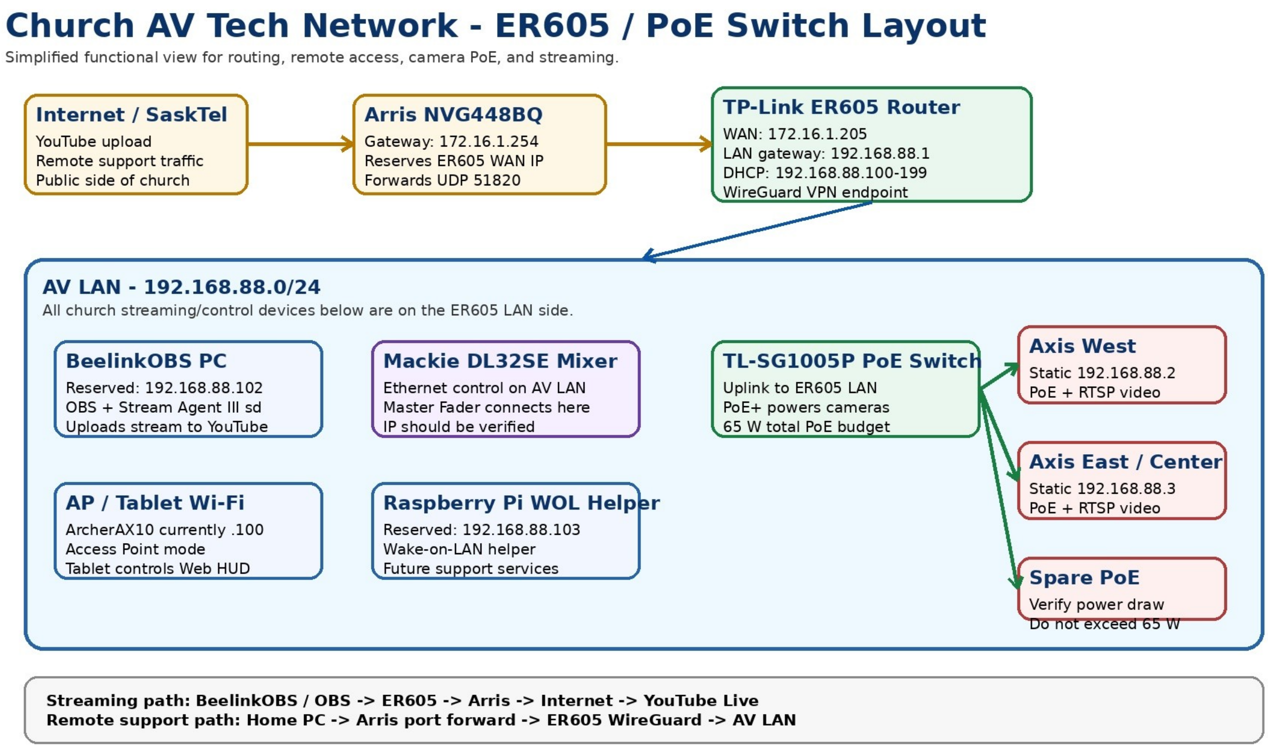

Figure 1 - Current AV network layout showing Arris -> ER605 -> AV LAN and PoE camera switch.

Main network roles

Device	Role in the AV system	Important address / note
Arris NVG448BQ	Main church internet gateway. Upstream router from SaskTel.	Gateway 172.16.1.254. Forwards WireGuard UDP 51820 to ER605 WAN.
TP-Link ER605	AV LAN router, DHCP server, firewall/NAT, and WireGuard VPN endpoint.	LAN 192.168.88.1. WAN currently 172.16.1.205.
TP-Link TL-SG1005P PoE switch	Powers Axis cameras and carries their Ethernet data.	4 PoE+ ports, 65 W total PoE budget.
BeelinkOBS PC	Streaming computer running OBS, Stream Agent III sd, Proclaim, Master Fader, and loopMIDI.	Reserved 192.168.88.102.
Axis West camera	West PTZ camera video/control.	Static 192.168.88.2.
Axis East/Center camera	East/Center PTZ camera video/control.	Static 192.168.88.3.
Raspberry Pi WOL helper	Always-on local helper for Wake-on-LAN and future support tasks.	Reserved 192.168.88.103.
Access Point / ArcherAX10	Wi-Fi access for tablet and mobile controls.	Currently seen at 192.168.88.100; should remain AP mode, not router mode.
Mackie DL32SE mixer	Audio mixer controlled by Master Fader over the AV LAN.	IP should be verified and reserved/static if needed.

4. Router access and login information

Credentials: Protect this section. It contains the ER605 login used for AV support.

Item	Value
ER605 web address from the AV LAN	https://192.168.88.1 or http://192.168.88.1
Router name / ID	NHCL
Router password	pjw3628
Normal access method	Use BeelinkOBS, an AV LAN computer, or WireGuard VPN remote access.
Browser warning	The browser may show Not secure. This is expected for a local router page without a public certificate.

5. Current ER605 configuration reference

5.1 System and WAN status

Setting	Current value / expected value
Hardware version shown in screenshots	ER605 v2.20
Firmware shown in screenshots	2.2.6 Build 20240718 Rel.82712
WAN connection type	Dynamic IP
WAN status	Link Up / Connected
WAN IP address	172.16.1.205
WAN subnet mask	255.255.255.0
WAN MAC address	A8-29-48-9A-C8-CF
WAN default gateway	172.16.1.254
Primary DNS	172.16.1.254

5.2 LAN and DHCP

Setting	Current value / expected value
LAN gateway IP	192.168.88.1
LAN subnet mask	255.255.255.0
DHCP server	Enabled
DHCP pool	192.168.88.100 through 192.168.88.199
Lease time	120 minutes
BeelinkOBS reservation	192.168.88.102 / MAC 78-55-36-03-3C-2E
Raspberry Pi reservation	192.168.88.103 / MAC B8-27-EB-EB-A0-FE
Axis camera addresses	192.168.88.2 and 192.168.88.3 are static camera-side addresses, outside DHCP pool.

5.3 Devices currently visible in DHCP client list

Device	IP address	Notes
BeelinkOBS	192.168.88.102	Permanent/reserved. Main streaming PC.
raspberrypi	192.168.88.103	Permanent/reserved. WOL helper.
Sound-s-Tab-A9	192.168.88.104	Tablet/client device; lease-based.
ArcherAX10	192.168.88.100	Access point; verify whether it should be static or reserved.

6. PoE switch and camera power design

The TP-Link TL-SG1005P is used because the ER605 does not provide PoE. The switch supplies both Ethernet data and PoE camera power to the Axis cameras, while the ER605 handles routing and VPN functions.

6.1 TL-SG1005P key specifications

Specification	Value
Ports	5 x 10/100/1000 Mbps RJ45 ports
PoE ports	Ports 1-4 provide PoE+
PoE standards	IEEE 802.3af/at compliant powered devices
Per-port PoE limit	Up to 30 W per PoE port
Total PoE budget	65 W total for all PoE ports
Switching capacity	10 Gbps
Packet forwarding rate	7.44 Mpps
Fan	Fanless
External power adapter	53.5 V DC / 1.31 A
Configuration	Unmanaged / no configuration required

The Axis M5525-E cameras are suitable for this PoE switch. The Axis datasheet lists PoE as IEEE 802.3af/802.3at Type 1 Class 3, typical 6.6 W and maximum 12.95 W. Two Axis cameras would therefore be under about 26 W maximum, leaving ample margin within the TL-SG1005P 65 W PoE budget. If a FoMaKo or any other PoE camera is reintroduced, verify that camera's PoE draw and keep the total below 65 W.

Comparison with MikroTik: The TL-SG1005P is not a managed router and does not replace the MikroTik feature-for-feature. In this design that is intentional: the ER605 handles router/VPN/DHCP functions, and the TL-SG1005P is a dedicated, standards-based PoE power/data switch for the cameras.

6.2 Why this is better for the current church setup

- Router and PoE power are separated, so a PoE power problem does not necessarily take down routing/VPN functions.
- Both ER605 and TL-SG1005P are compact, low-power, fanless devices that fit well in the AV tech cabinet.
- The equipment is physically accessible for troubleshooting, unlike the earlier higher-shelf router location.
- Shorter Ethernet runs in the AV tech cabinet reduce cable complexity and make visual inspection easier.
- PoE budget is sufficient for the current Axis-only camera setup with significant headroom.

7. WireGuard VPN access overview

WireGuard is the remote access path into the AV LAN. It lets the home/support computer reach BeelinkOBS, the ER605, cameras, and other AV devices without exposing those devices directly to the internet.

Item	Current reference value / note
ER605 WireGuard listen port	UDP 51820
Arris port forward	UDP 51820 -> ER605 WAN 172.16.1.205
Client endpoint	Church public address on port 51820; verify current public IP before relying on it.
ER605 tunnel-side address	10.6.0.1
Remote client address	10.6.0.2/32
Client AllowedIPs	192.168.88.0/24 and 10.6.0.1/32
Client MTU	1280 has been used successfully in testing.
Keys	Do not publish private keys. This document intentionally does not print private keys.

Key handling: The uploaded ER605 screenshot set did not include a WireGuard key page. If future screenshots include private keys or QR codes, redact them before adding the images to the public/shared documentation set.

8. Configuration procedures

8.1 Physical installation

1. Connect the church Arris LAN feed to the ER605 WAN port.
2. Connect ER605 LAN ports to the AV LAN devices, access point, and/or the PoE switch uplink.
3. Connect Axis cameras to PoE ports 1-4 on the TL-SG1005P. Use port 5 or any non-PoE/uplink arrangement as appropriate for the ER605 uplink.
4. Keep the access point in Access Point mode. Do not allow it to run a second DHCP server.
5. Place the ER605 and PoE switch in the AV tech cabinet for easier access and shorter cabling.

8.2 Verify ER605 status after power-up

6. From BeelinkOBS or another AV LAN computer, browse to https://192.168.88.1.
7. Log in using the ER605 credentials in this document.
8. Open Status -> System Status.
9. Confirm the WAN is Link Up and that the WAN IP is still 172.16.1.205.
10. Confirm the LAN gateway remains 192.168.88.1.

8.3 WAN setup

11. Open Network -> WAN.
12. Confirm WAN Mode is using the WAN port as the active WAN connection.
13. Confirm the WAN connection type is Dynamic IP.
14. Do not manually set the ER605 WAN IP unless there is a planned change. The Arris DHCP reservation should keep the WAN IP stable.

8.4 LAN and DHCP setup

15. Open Network -> LAN.
16. Confirm the LAN network is 192.168.88.1 / 255.255.255.0.
17. Confirm DHCP Server is enabled.
18. Confirm the DHCP range is 192.168.88.100 through 192.168.88.199.
19. Use Address Reservation for devices that must stay predictable, especially BeelinkOBS and the Raspberry Pi WOL helper.
20. Keep static camera IPs such as 192.168.88.2 and 192.168.88.3 outside the DHCP pool.

8.5 Arris-side requirements

These settings are on the upstream Arris router, not on the ER605. They are included here because WireGuard access depends on them.

- The Arris should reserve the ER605 WAN MAC A8-29-48-9A-C8-CF to 172.16.1.205.
- The Arris should forward UDP 51820 to 172.16.1.205.
- If the ER605 WAN IP changes, remote WireGuard access can fail until the reservation and port forward match again.

8.6 Backup after known-good changes

21. After any successful router configuration change, export or back up the ER605 configuration.
22. Store the backup with the AV tech documentation, but keep it protected because it can contain sensitive network settings.
23. Record the date and what changed so the next support person knows whether the backup is current.

9. Troubleshooting and recovery

Symptom	Most likely checks
Cannot open ER605 page at 192.168.88.1	Confirm PC is on the AV LAN; ping 192.168.88.1; check cable to ER605 LAN port; verify Wi-Fi is on the AV access point.
WireGuard does not connect from home	Check church internet, Arris port forward UDP 51820, ER605 WAN IP 172.16.1.205, ER605 WireGuard service/listen port, client endpoint, keys, and MTU.
WireGuard connects but pages do not load	Confirm client AllowedIPs include 192.168.88.0/24; verify MTU 1280; ping 192.168.88.1; test browser after ping succeeds.
Camera video or PTZ control fails	Check PoE switch power, camera link LEDs, camera IPs 192.168.88.2/.3, ER605 LAN cabling, and whether the camera web page opens.
BeelinkOBS changes IP	Check ER605 Address Reservation for MAC 78-55-36-03-3C-2E -> 192.168.88.102.
Raspberry Pi WOL helper changes IP	Check ER605 Address Reservation for MAC B8-27-EB-EB-A0-FE -> 192.168.88.103.
Master Fader / mixer unstable	Check mixer Ethernet cable, ER605/AP links, IP addressing, and whether other AV LAN devices remain stable.
Multiple DHCP servers suspected	Confirm the access point is in AP mode and does not run DHCP. ER605 should be the only DHCP server on 192.168.88.x.

10. Screenshots and reference figures

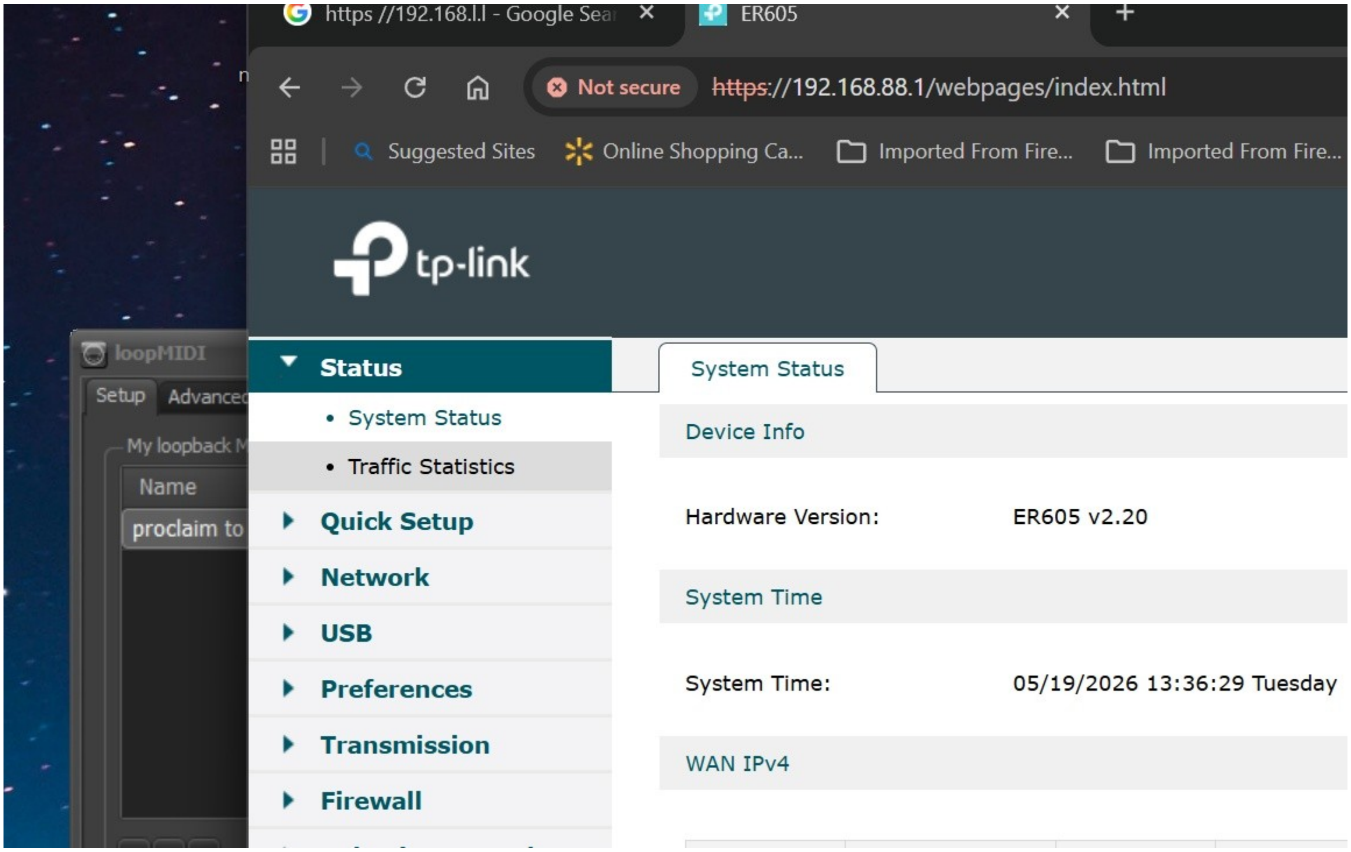

Figure 2 - ER605 Status page showing hardware/firmware and WAN IPv4 status.

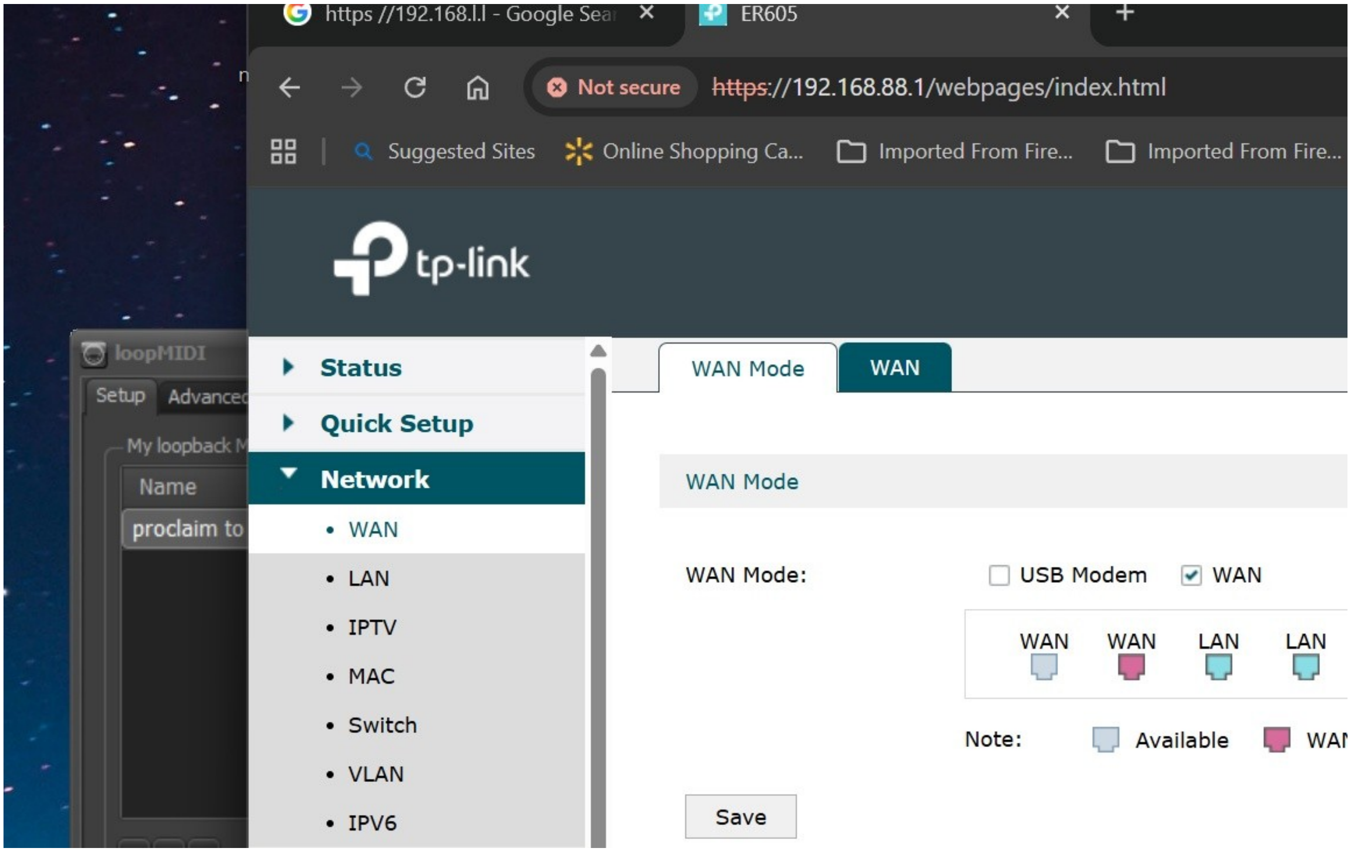

Figure 3 - Network -> WAN Mode page showing the WAN port selected as the WAN connection.

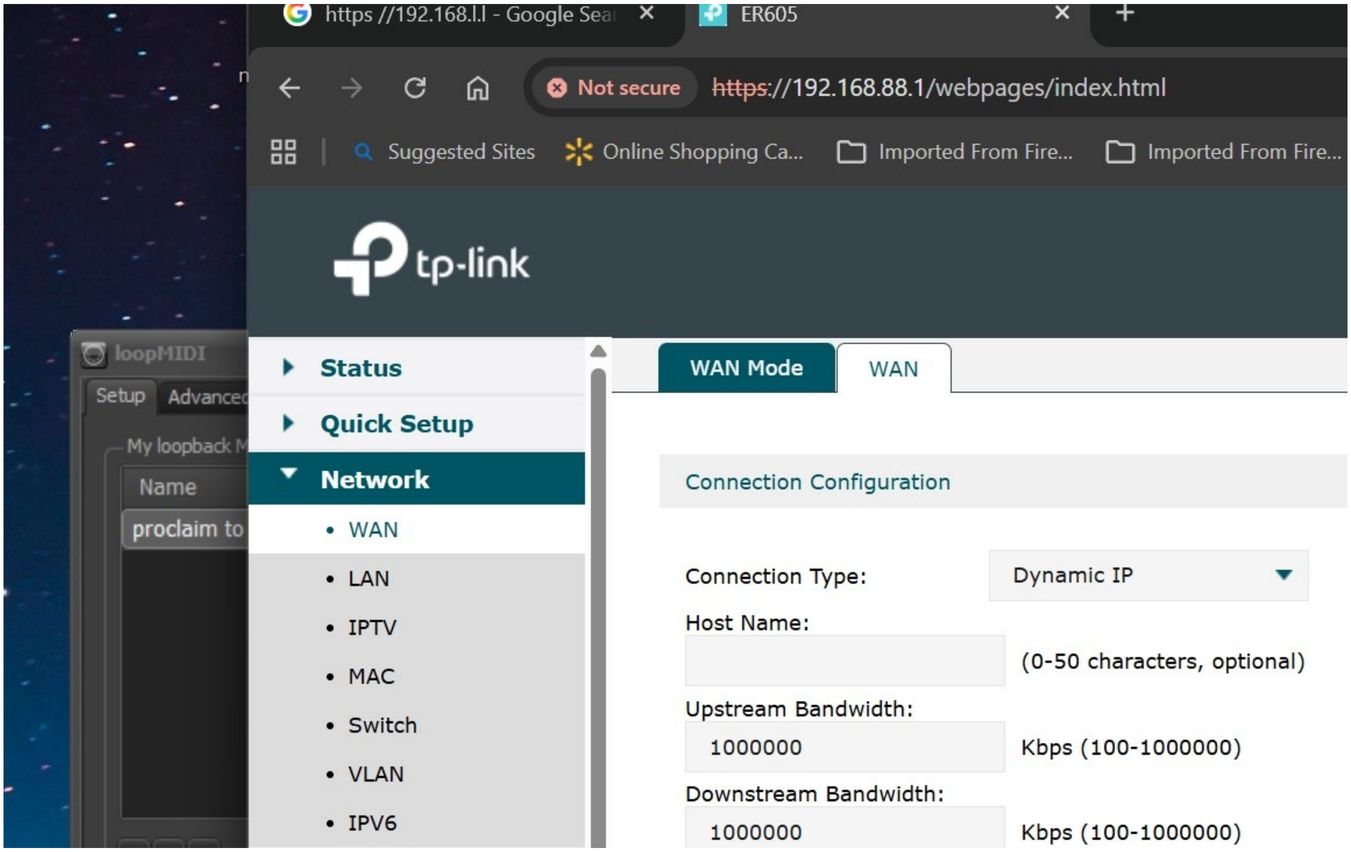

Figure 4 - Network -> WAN page showing Dynamic IP and the current WAN status.

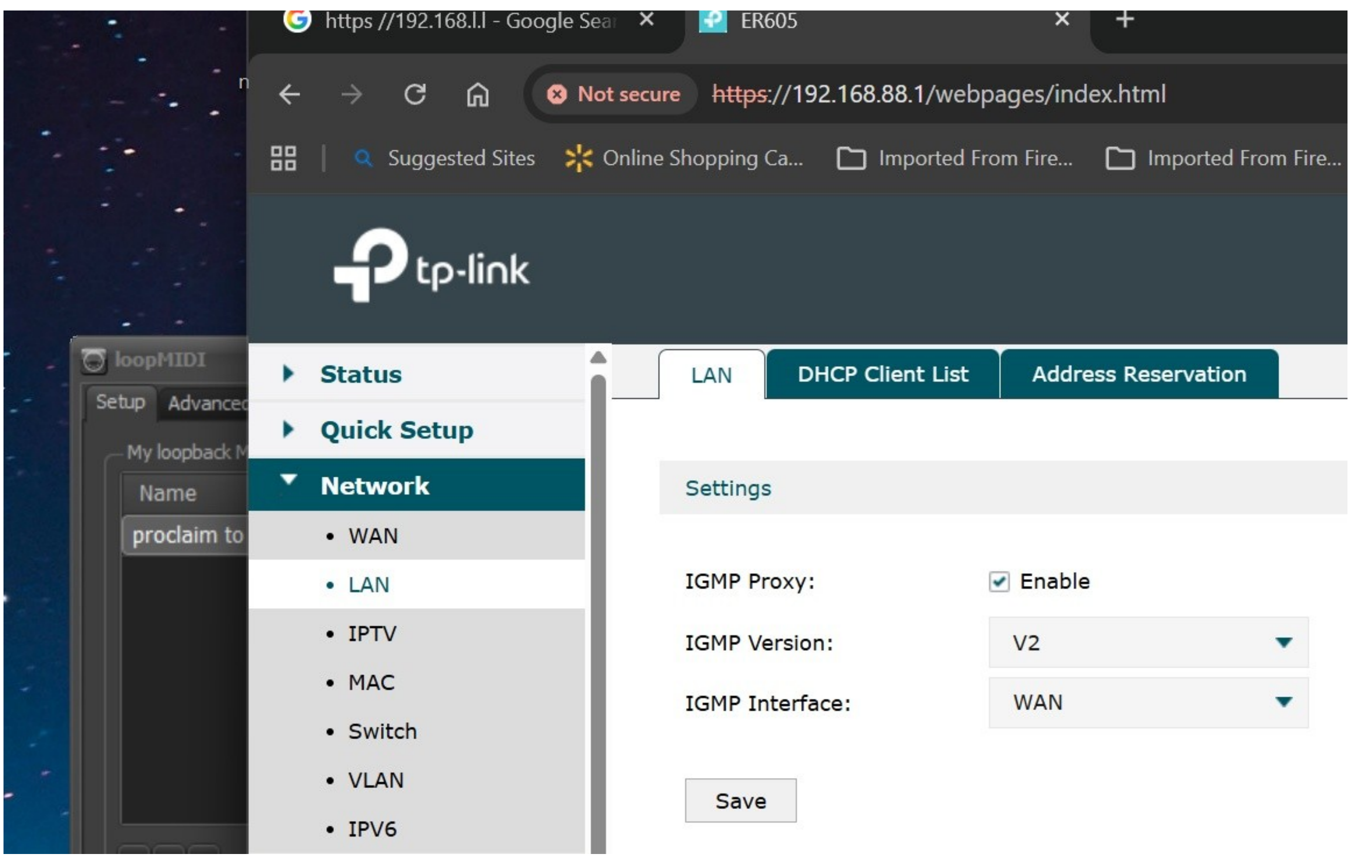

Figure 5 - Network -> LAN summary showing LAN 192.168.88.1/24 and DHCP enabled.

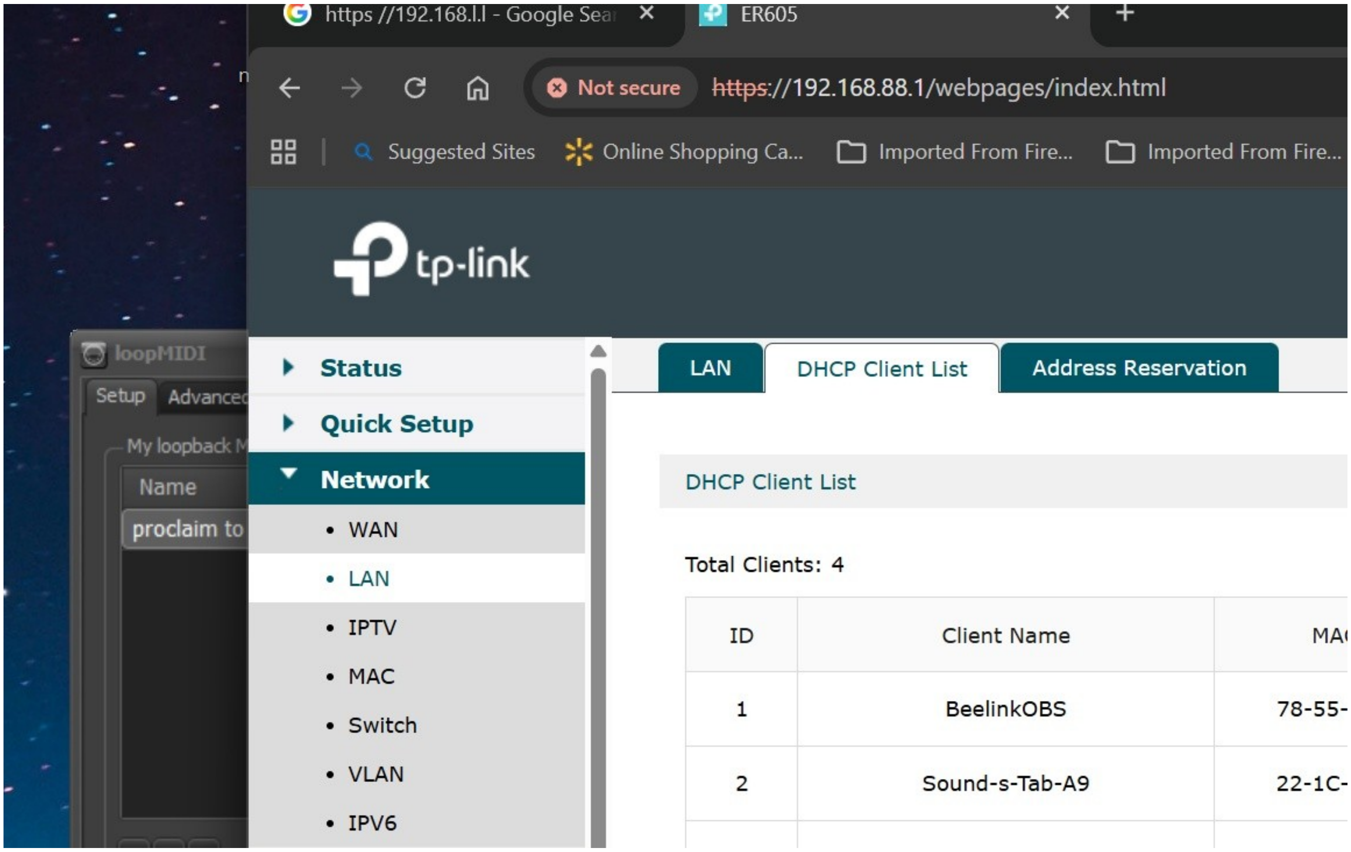

Figure 6 - DHCP Client List showing BeelinkOBS, Raspberry Pi, tablet, and ArcherAX10.

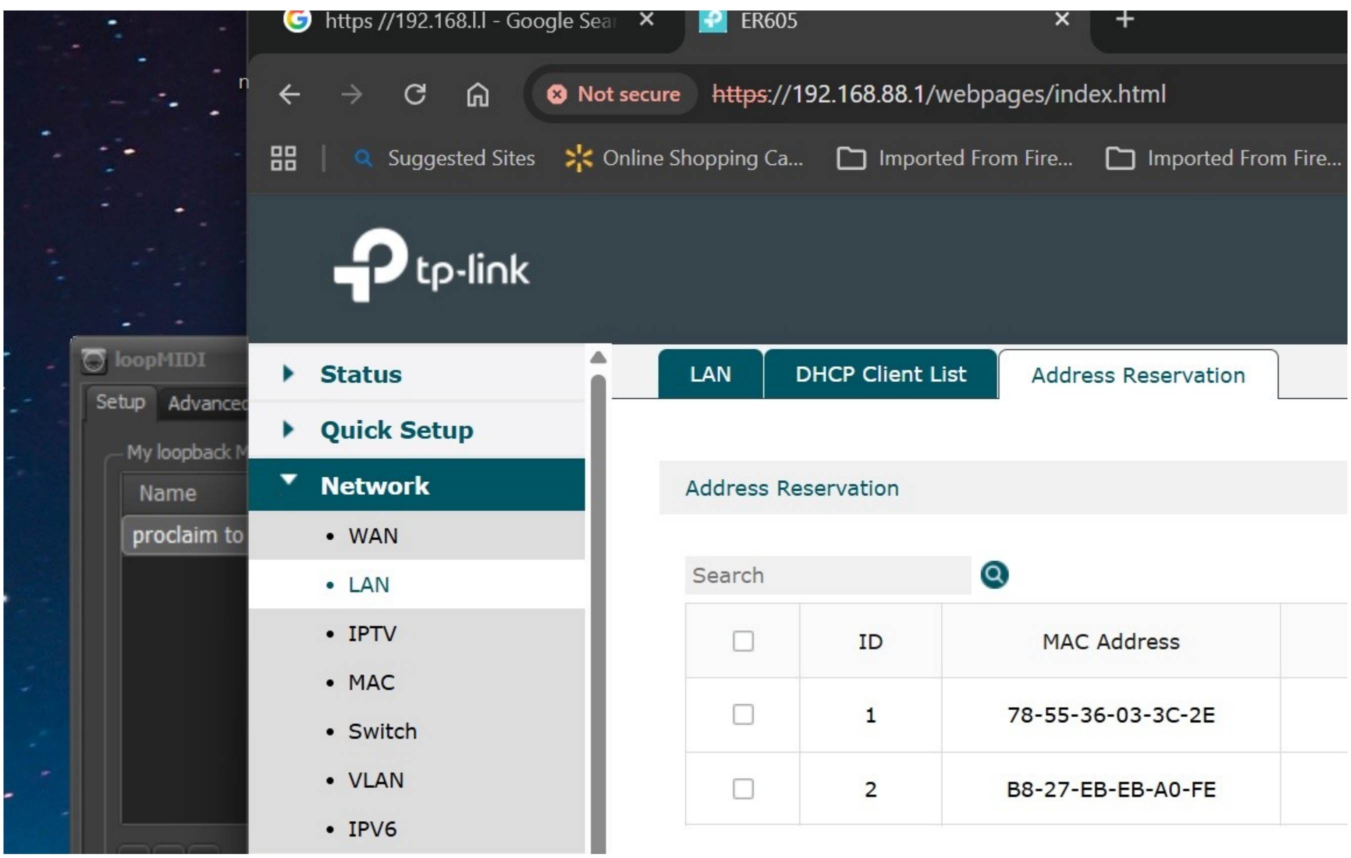

Figure 7 - Address Reservation page showing reserved BeelinkOBS and Raspberry Pi addresses.

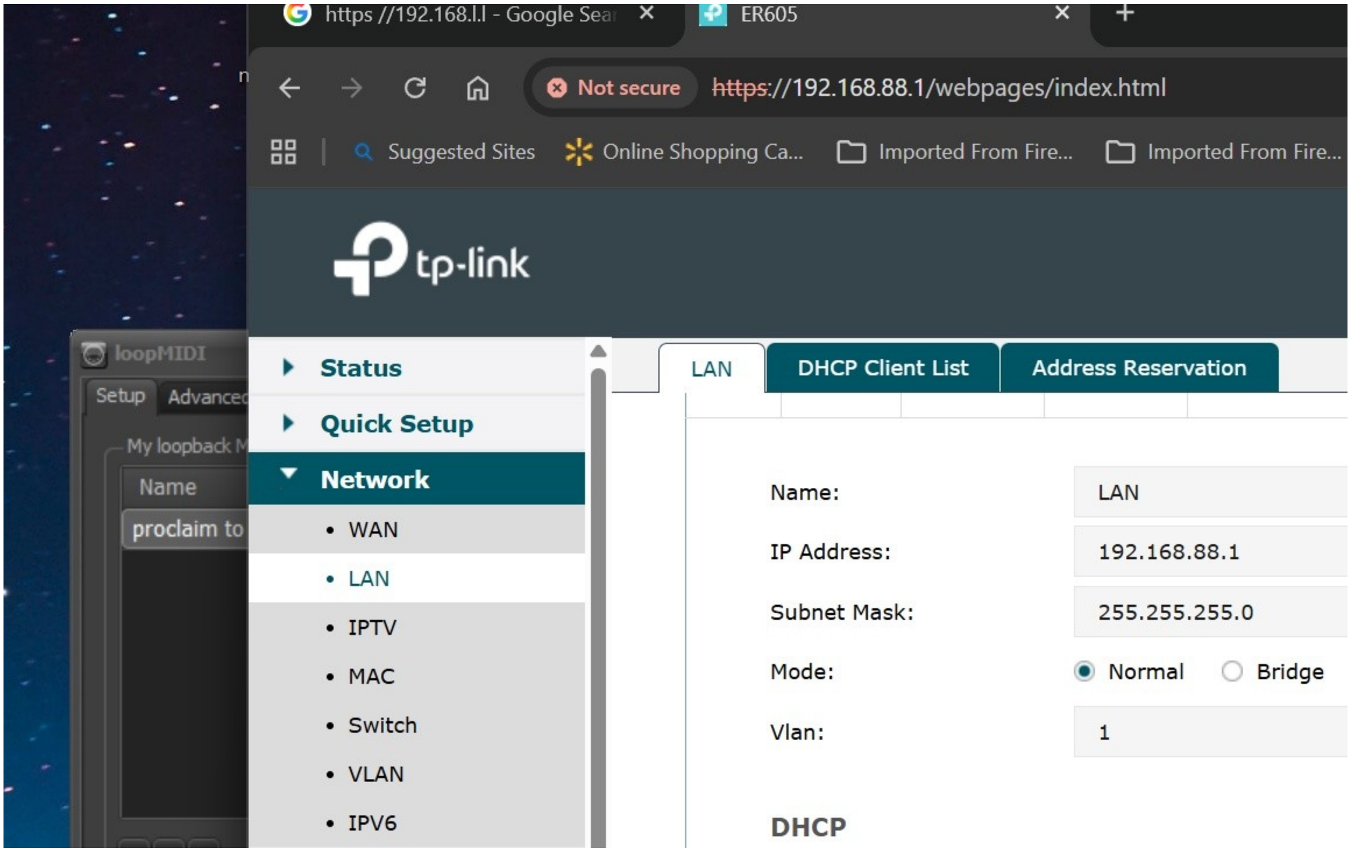

Figure 8 - LAN/DHCP detail page showing DHCP pool 192.168.88.100-199.

11. Source notes for hardware specifications

Hardware specification references used for this guide:

- TP-Link Canada, TL-SG1005P V2 product specifications: 5 gigabit ports, ports 1-4 PoE+, 30 W per PoE port, 65 W total PoE budget, fanless, 10 Gbps switching capacity.
- TP-Link Canada, ER605 V2 product specifications: five gigabit ports, multi-WAN capability, VPN router features, WireGuard VPN throughput listed at 123.4 Mbps, NAT throughput near gigabit class in TP-Link tests.
- Axis M5525-E datasheet: Power over Ethernet IEEE 802.3af/802.3at Type 1 Class 3, typical 6.6 W, maximum 12.95 W.